

TRANSFORMER ARCHITECTURE

Vincent Guigue,
inspiré des supports de Benjamin Piwowarski & Thomas G rald

INTRODUCTION



Aggregation problem

Sequence : $\mathbf{s} = \{s_1, \dots, s_L\}$

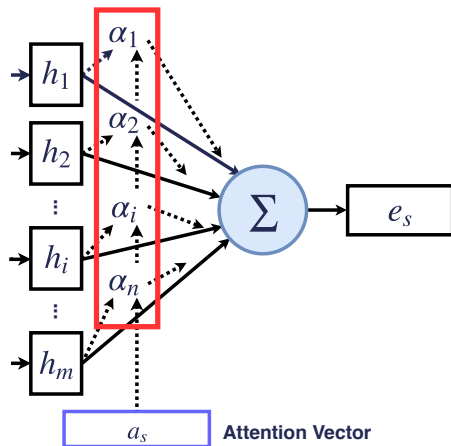
Representation learning : $s_i \rightarrow \mathbf{e}_i \in \mathbb{R}^Z$

Decision on $\mathbf{s} \Rightarrow$ Representation / mapping $R(\mathbf{s}) : \mathbb{R}^{L \times Z} \mapsto \mathbb{R}^d$

+ Decision function $f(R(\mathbf{s})) : \mathbb{R}^d \mapsto \mathcal{Y}$

- RNN + last hidden layer (ext: Bi-RNN)
- CNN/RNN + pooling
- CNN/RNN + attention = learnt weighed sum

Attention & aggregation vs Diffusion process



- Learning to weight localized representation
- Many efficient applications [2014-2018]
- Few parameters / Sparsity
- Attention vector = Query vector
- How to code it in pytorch?

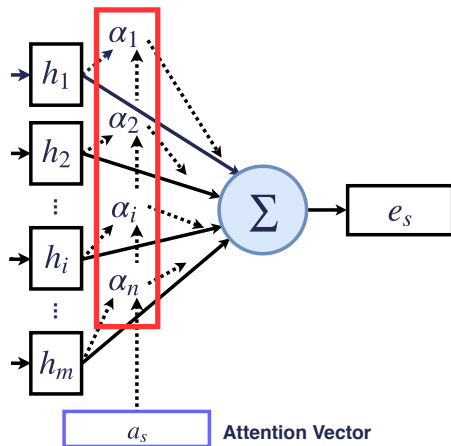
Practical Session:

Q1 Preliminary tests: uniform attention

Q2 Classical global attention



Attention & aggregation vs Diffusion process



- Learning to weight localized representation
- Many efficient applications [2014-2018]
- Few parameters / Sparsity
- Attention vector = Query vector
- How to code it in pytorch?

Q3 $\Rightarrow$ Reconstruction of **all embedding** by recombination = diffusion process



Code for attention pooling

Imagine chaining a new module after a RNN

```
1 import torch
2 import torch.nn as nn
3 import torch.nn.functional as F
4
5 class AttentionPooling(nn.Module):
6     def __init__(self, dim):
7         self.attn = nn.Linear(dim, 1, bias=False)
8
9     def forward(self, h):          # h: (B, T, Z) Batch (first) x Time x LatentDim
10         scores = self.attn(h)      # (B, T, 1)
11         weights = F.softmax(scores, dim=1) # (B, T, 1)
12
13         pooled = (weights * h).sum(dim=1) # (B, Z)
14         return pooled
```

- Very simple code...
- ... And automatic differentiation & batch compatibility (using Linear module)



Inspiration from memory networks (question answering)

Task:

Sam walks into the kitchen.
Sam picks up an apple.
Sam walks into the bedroom.
Sam drops the apple.

Q: Where is the apple?

A. Bedroom

Brian is a lion.
Julius is a lion.
Julius is white.
Bernhard is green.

Q: What color is Brian?

A. White

Mary journeyed to the den.
Mary went back to the kitchen.
John journeyed to the bedroom.
Mary discarded the milk.

Q: Where was the milk before the den?

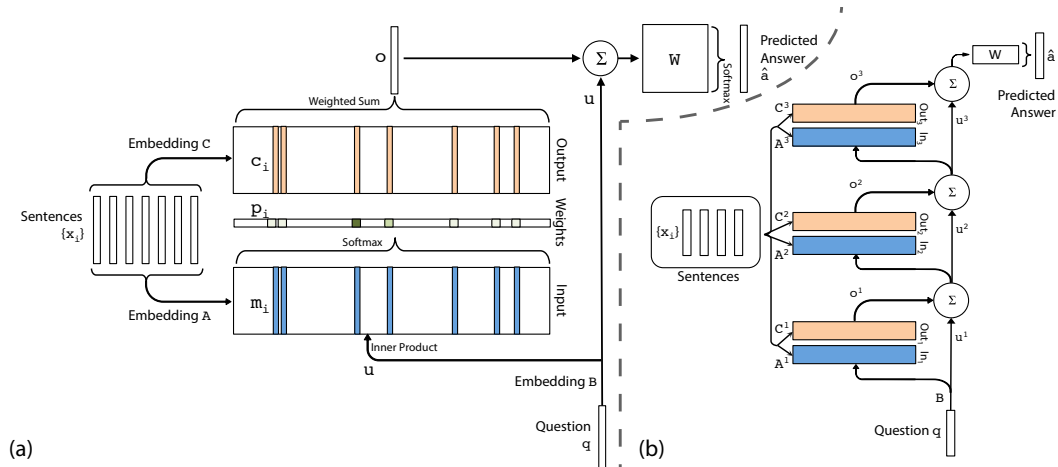
A. Hallway

Babi dataset

How to anchor question answering in Knowledge Bases?

Inspiration from memory networks (question answering)

How to anchor question answering in Knowledge Bases?



Inspiration from memory networks (question answering)

How to anchor question answering in Knowledge Bases?

Story (1: 1 supporting fact)	Support	Hop 1	Hop 2	Hop 3
Daniel went to the bathroom.		0.00	0.00	0.03
Mary travelled to the hallway.		0.00	0.00	0.00
John went to the bedroom.		0.37	0.02	0.00
John travelled to the bathroom.	yes	0.60	0.98	0.96
Mary went to the office.		0.01	0.00	0.00
Where is John? Answer: bathroom Prediction: bathroom				

Story (16: basic induction)	Support	Hop 1	Hop 2	Hop 3
Brian is a frog.	yes	0.00	0.98	0.00
Lily is gray.		0.07	0.00	0.00
Brian is yellow.	yes	0.07	0.00	1.00
Julius is green.		0.06	0.00	0.00
Greg is a frog.	yes	0.76	0.02	0.00
What color is Greg? Answer: yellow Prediction: yellow				

Story (2: 2 supporting facts)	Support	Hop 1	Hop 2	Hop 3
John dropped the milk.		0.06	0.00	0.00
John took the milk there.	yes	0.88	1.00	0.00
Sandra went back to the bathroom.		0.00	0.00	0.00
John moved to the hallway.	yes	0.00	0.00	1.00
Mary went back to the bedroom.		0.00	0.00	0.00
Where is the milk? Answer: hallway Prediction: hallway				

Story (18: size reasoning)	Support	Hop 1	Hop 2	Hop 3
The suitcase is bigger than the chest.	yes	0.00	0.88	0.00
The box is bigger than the chocolate.		0.04	0.05	0.10
The chest is bigger than the chocolate.	yes	0.17	0.07	0.90
The chest fits inside the container.		0.00	0.00	0.00
The chest fits inside the box.		0.00	0.00	0.00
Does the suitcase fit in the chocolate? Answer: no Prediction: no				



Conclusion

Current limitations with RNN:

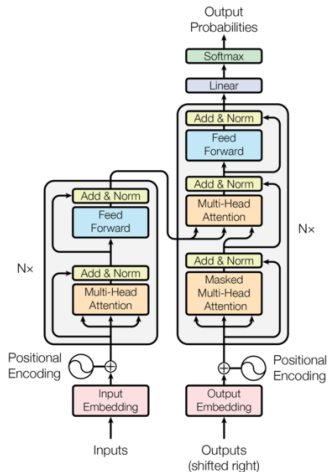
- Long dependencies
- Global message representation
 - Continuous indexing
- Targeted dependencies
 - Co-reference resolution

ARCHITECTURE

Transformer generality

Several sources to understand Transformers [easier than the original article]

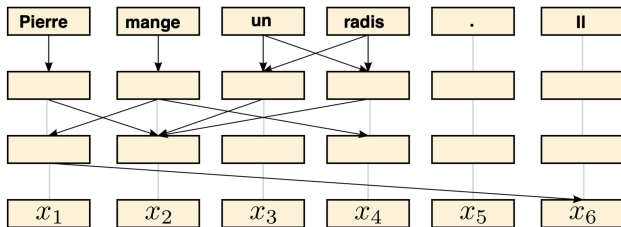
- J. Alammar 2018 <http://jalammar.github.io/illustrated-transformer/>
- Alexander Rush 2018 <http://nlp.seas.harvard.edu/2018/04/03/attention.html>
- P. Bloem 2019 <http://www.peterbloem.nl/blog/transformers>



Overcoming classical pictures
⇒ re-implementing a transformer block

Subword representation

General idea: representation diffusion + learning the diffusion



Scale choice

Character

- + (very) small vocabulary
- no semantics

Word

- + Semantics
- ++
- Huge vocabulary
- Unknown words

 Sennrich, R. et al. 2016. *Neural Machine Translation of Rare Words with Subword Units*.

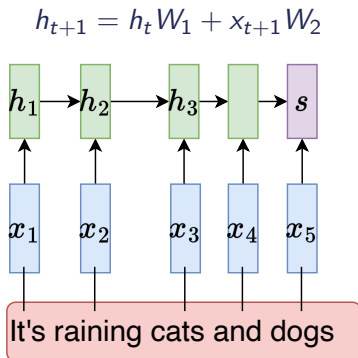
 **les chats sont des félinés**

est transformé en

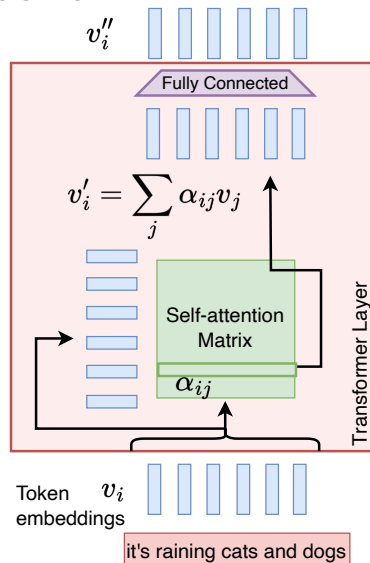
[CLS] _les _chats _sont _des _f éli dés [SEP]

Transformer architecture: state-of-the-art aggregation

Recurrent Neural Network:



Transformer:

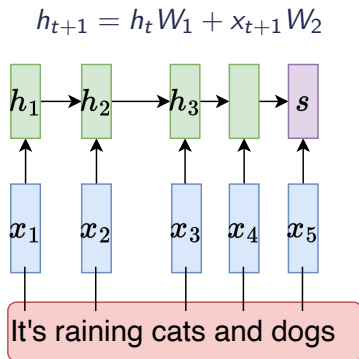


Attention is all you need, [Vaswani et al. NeurIPS 2017](#)

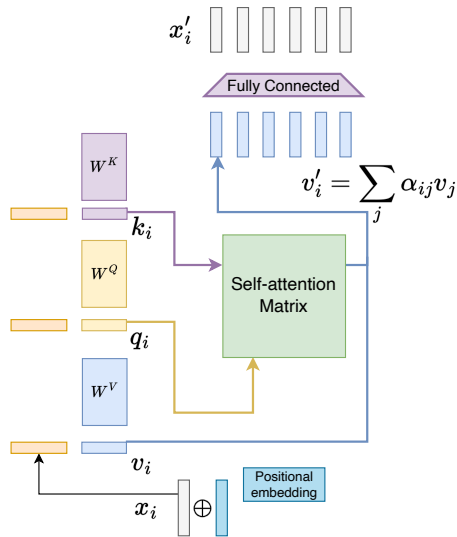
Sequence to Sequence Learning with Neural Networks, [Sutskever et al. NeurIPS 2014](#)

Transformer architecture: state-of-the-art aggregation

Recurrent Neural Network:



Transformer:



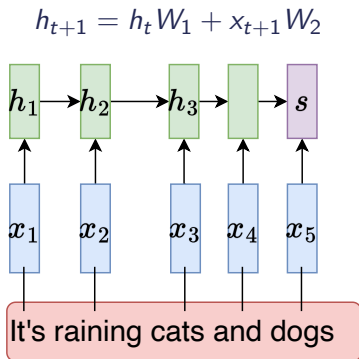
Attention is all you need, [Vaswani et al. NeurIPS 2017](#)

Sequence to Sequence Learning with Neural Networks, [Sutskever et al. NeurIPS 2014](#)

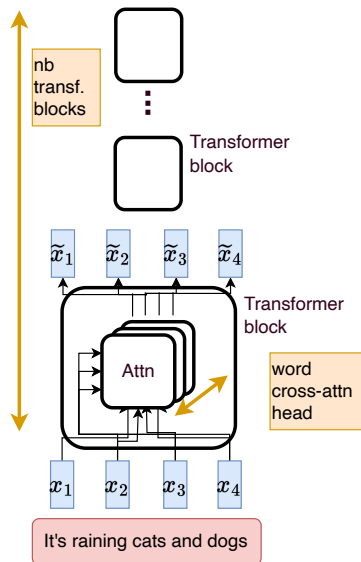
the cat sat on the mat

Transformer architecture: state-of-the-art aggregation

Recurrent Neural Network:



Transformer:

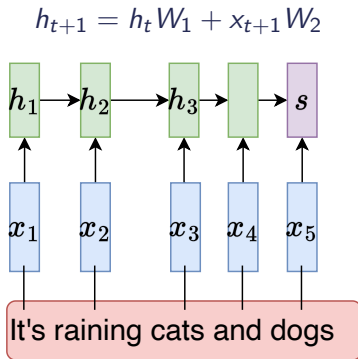


Attention is all you need, [Vaswani et al. NeurIPS 2017](#)

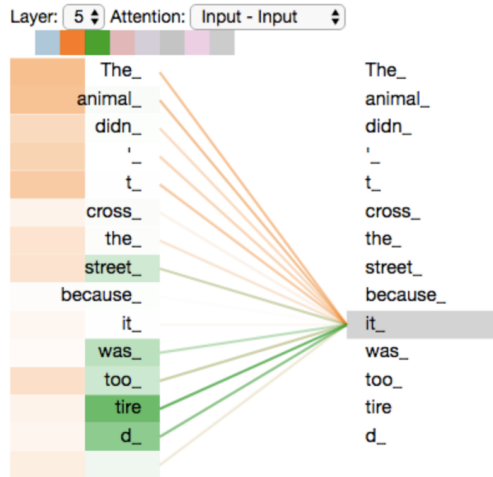
Sequence to Sequence Learning with Neural Networks, [Sutskever et al. NeurIPS 2014](#)

Transformer architecture: state-of-the-art aggregation

Recurrent Neural Network:



Transformer:



Attention is all you need, [Vaswani et al. NeurIPS 2017](#)

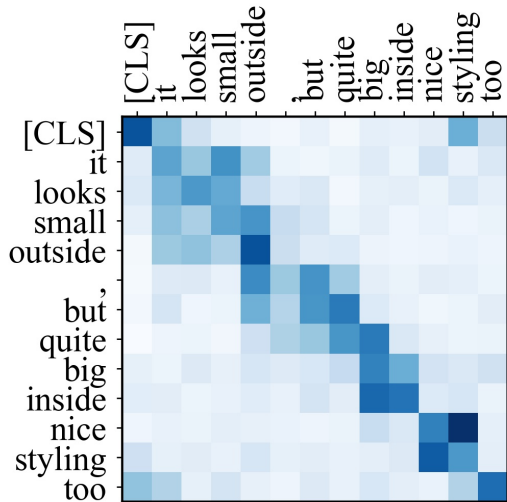
Sequence to Sequence Learning with Neural Networks, [Sutskever et al. NeurIPS 2014](#)

Attention block: General idea

- 1 Computing self-attention between tokens
- 2 Normalizing attention (softmax)
- 3 Exploiting attention to recombine token representation

⇒ How to compute interactions
(=self attention)

⇒ New token representation
(=contextual representation)

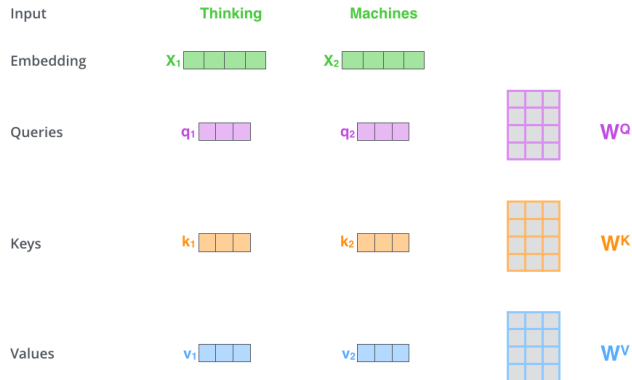


Attention block

Interactions through vectors derived from the embedding

1 Compute **Keys**, **Queries**, **Values** by linear mapping

Note: here are the main parameters of the transformer architecture



Attention block

Computing the attention

- 1 Inner product between keys/queries
- 2 Division by key/query size
 - Softmax too sensitive to large values
 - Temperature
- 3 Softmax $\Rightarrow$ over the relevant dimension

Input

Embedding

Queries

Keys

Values

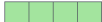
Score

Divide by 8 ($\sqrt{d_k}$)

Softmax

Thinking

Machines

 x_1  x_2  q_1  q_2  k_1  k_2  v_1  v_2  $q_1 \cdot k_1 = 112$ $q_1 \cdot k_2 = 96$

14

12

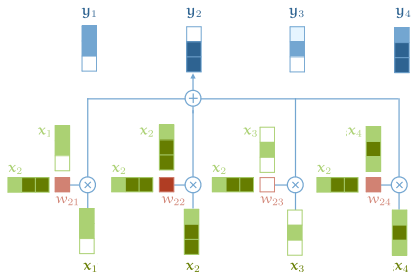
0.88

0.12



Attention block

New values:



$$v_i^{\ell+1} = \sum \alpha_k v_k^{\ell}$$

How to come back to the
embedding space?
value $\mapsto$ embedding

Input

Embedding

Queries

Keys

Values

Score

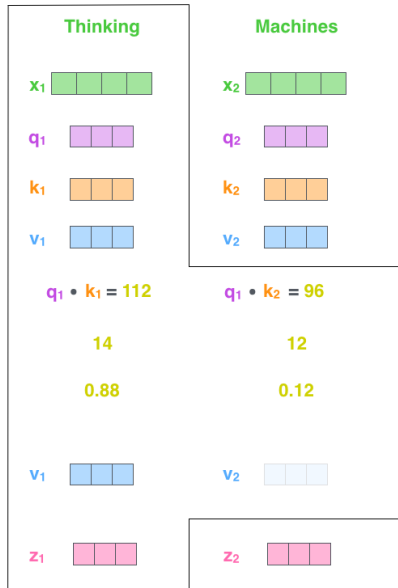
Divide by 8 ($\sqrt{d_k}$)

Softmax

Softmax

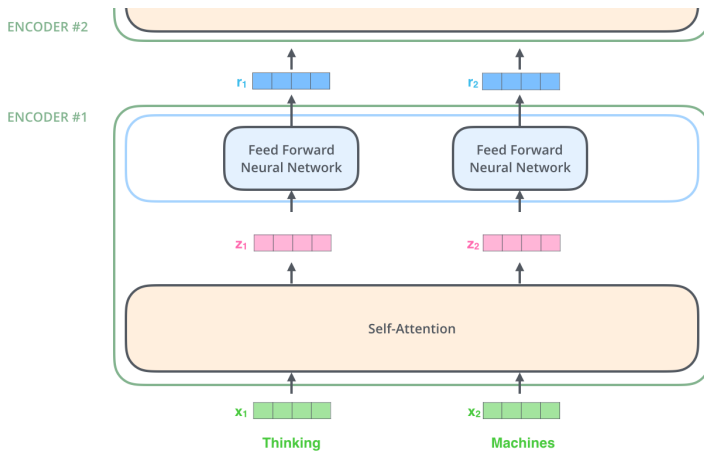
X
Value

Sum



Attention block

Use a fully connected to map to embedding space





Parallelism $\Rightarrow$ Batch of data

All tokens at once !

$$\begin{matrix} \text{X} \\ \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \end{matrix} \times \begin{matrix} \text{W}^{\text{Q}} \\ \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \end{matrix} = \begin{matrix} \text{Q} \\ \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \end{matrix}$$

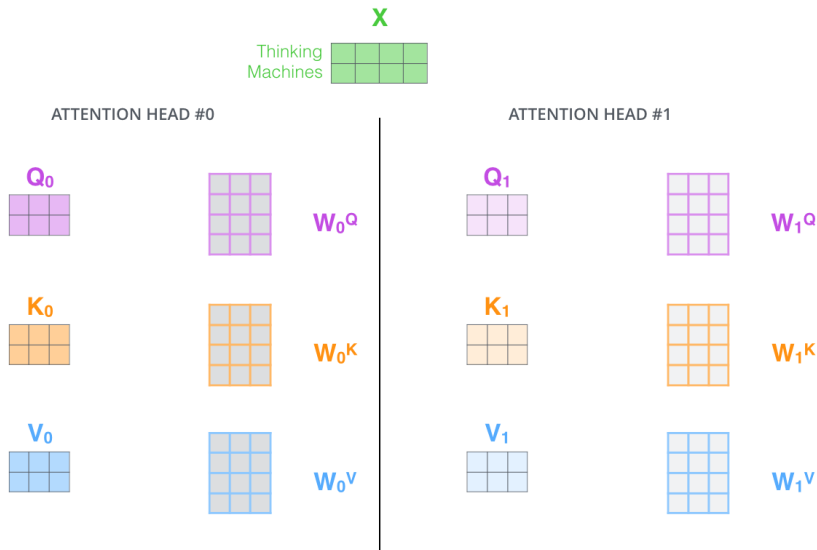
$$\begin{matrix} \text{X} \\ \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \end{matrix} \times \begin{matrix} \text{W}^{\text{K}} \\ \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \end{matrix} = \begin{matrix} \text{K} \\ \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \end{matrix}$$

$$\begin{matrix} \text{X} \\ \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \end{matrix} \times \begin{matrix} \text{W}^{\text{V}} \\ \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \end{matrix} = \begin{matrix} \text{V} \\ \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \end{matrix}$$

$$\begin{aligned} & \text{softmax} \left(\frac{\begin{matrix} \text{Q} \\ \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \end{matrix} \times \begin{matrix} \text{K}^{\text{T}} \\ \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \end{matrix}}{\sqrt{d_k}} \right) \begin{matrix} \text{V} \\ \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \end{matrix} \\ & = \begin{matrix} \text{Z} \\ \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \end{matrix} \end{aligned}$$

Multiple heads : dimension compatibility challenge

Why only **one** interaction?



Multiple heads : dimension compatibility challenge

Why only **one** interaction?

1) This is our input sentence*

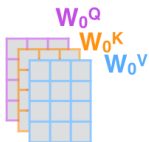
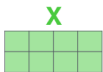
2) We embed each word*

3) Split into 8 heads.
We multiply **X** or **R** with weight matrices

4) Calculate attention using the resulting **Q**/**K**/**V** matrices

5) Concatenate the resulting **Z** matrices, then multiply with weight matrix **W^o** to produce the output of the layer

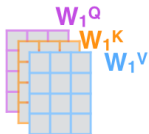
Thinking Machines



W^o



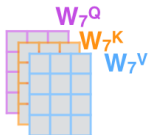
* In all encoders other than #0, we don't need embedding. We start directly with the output of the encoder right below this one



...

...

...



Multiple heads : dimension compatibility challenge

Why only **one** interaction?

Typically, values are concatenated then mapped to rebuild embeddings

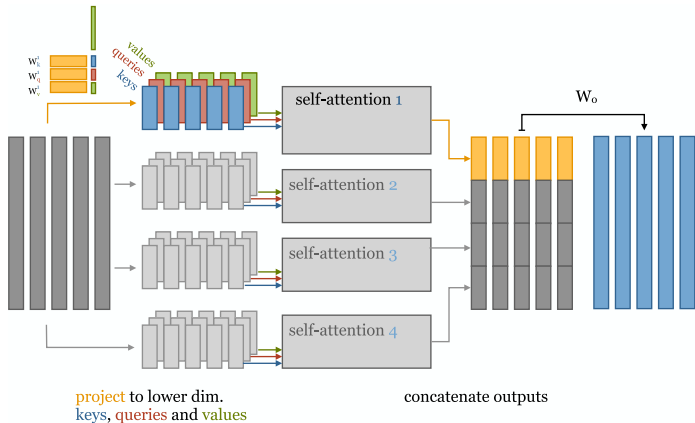
BERT

- Embedding : 768
- key/query/values : 64
- ... & 12 heads

$$\Rightarrow 12 \times 64 = 768$$

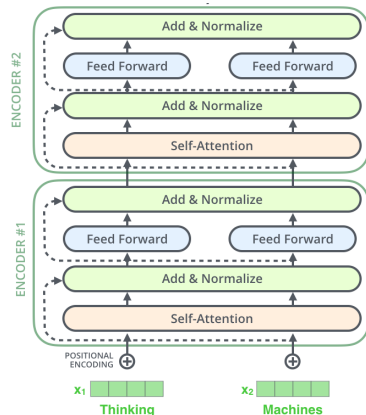
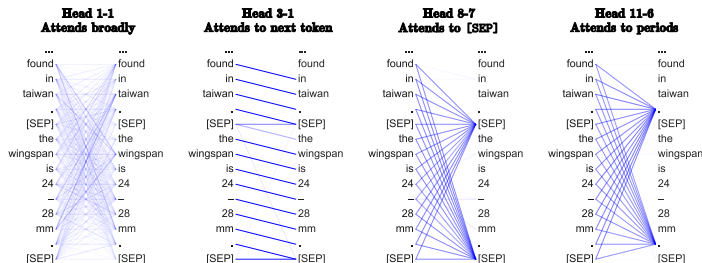
ChatGPT:

- Embedding : 12288
- key/query/values : 128
- 96 heads



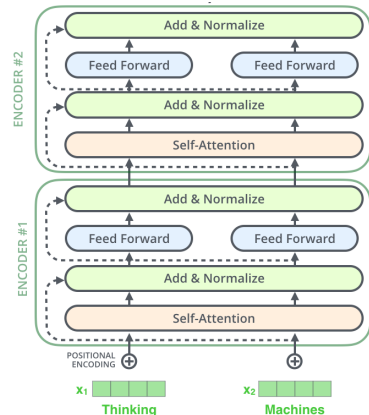
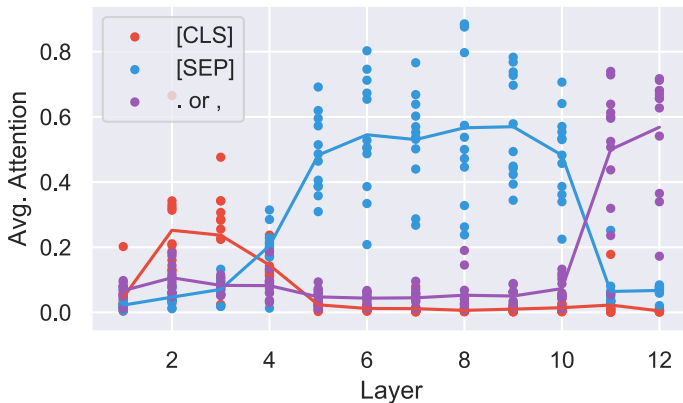
Multiple layers

- Have a look on the skip connexion
- Diffusion is made step-by-step
- Bertology shows that different relations are extracted at different scales



Multiple layers

- Have a look on the skip connexion
- Diffusion is made step-by-step
- Bertology shows that different relations are extracted at different scales

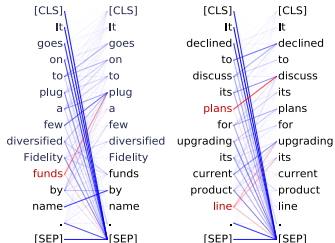


Multiple layers

- Have a look on the skip connexion
- Diffusion is made step-by-step
- Bertology shows that different relations are extracted at different scales

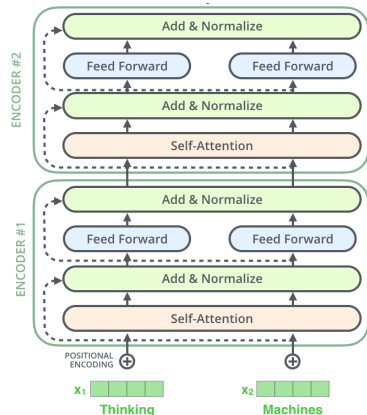
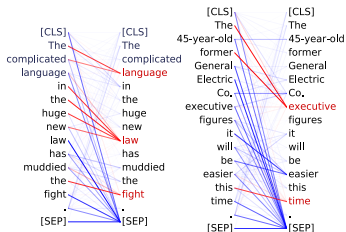
Head 8-10

- **Direct objects** attend to their verbs
- 86.8% accuracy at the dobj relation



Head 8-11

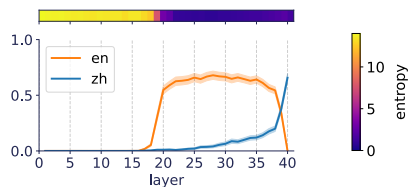
- **Noun modifiers** (e.g., determiners) attend to their noun
- 94.3% accuracy at the det relation



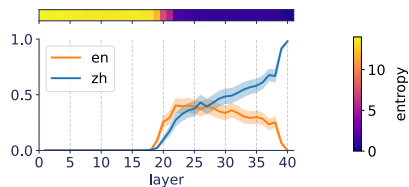
Multiple layers (& languages)

- Translate from French to Chinese
(+ rephrase/repeat Chinese + finish the sentence Chinese)
- Try to find existing nearby embedding
- Identify language

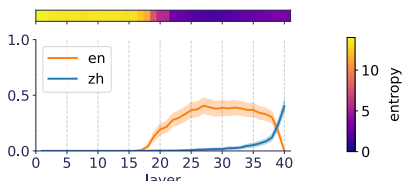
(a) Translation task



(b) Repetition task



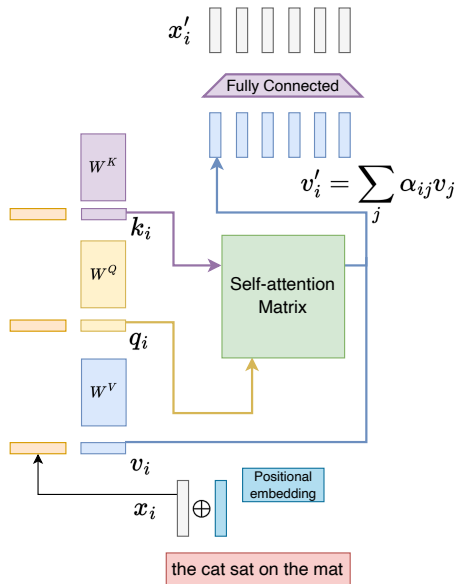
(c) Cloze task



[Wendler et al. 2024] Do Llamas Work in English?
On the Latent Language of Multilingual Transformers

Position coding

Until now, Transformer does not take position into account !!



Position coding

Until now, Transformer does not take position into account !!



$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{size}}}\right)$$

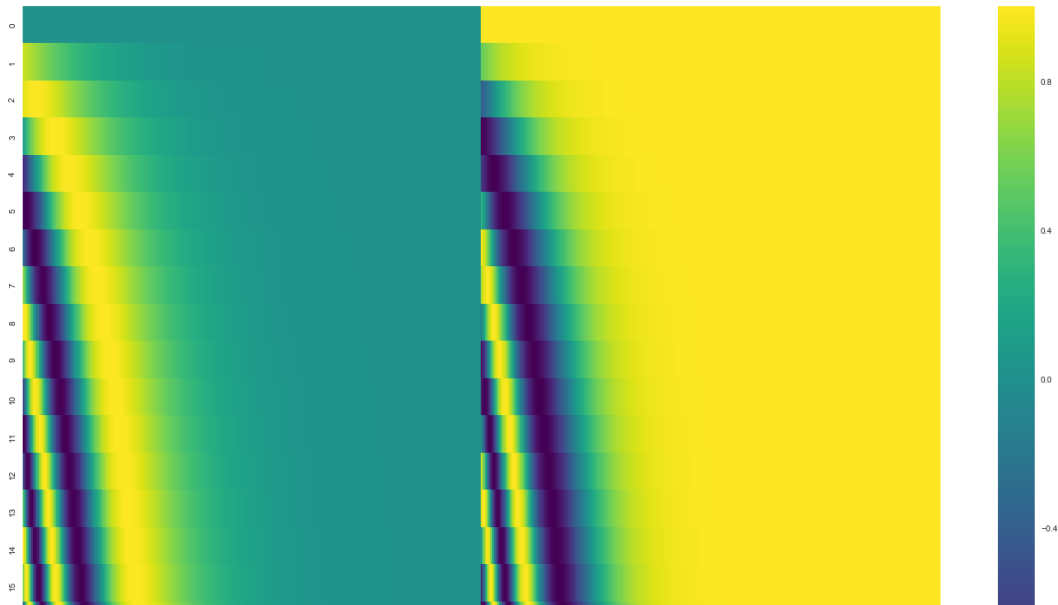
$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10000^{\frac{2i}{size}}}\right)$$

pos : position
 i : index of latent dim
 $size$: latent dim



Position coding

Until now, Transformer does not take position into account !!



Masking attention computation

Think about **batches** :

Padded sequences						Length
						4
						1
						6
						3

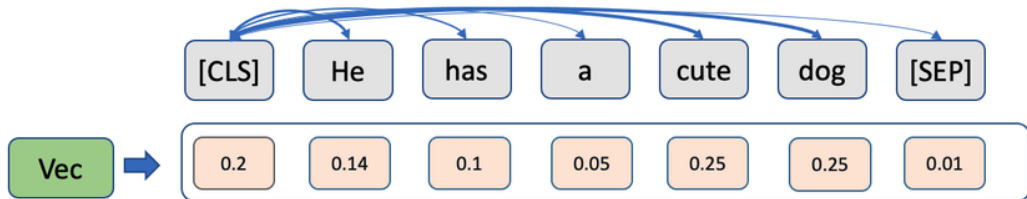
- Lengths are different
- Attention for PAD token = $-\infty \Rightarrow \alpha_{PAD} = 0$
Padding should not influence other tokens



Specific tokens

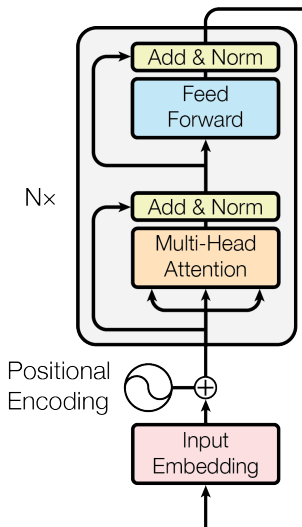
How to split texts or aggregate all informations? $\Rightarrow$ Introducing specific tokens:

- CLS
- SEP



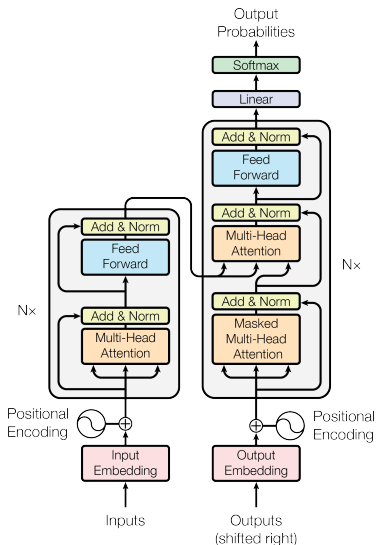
Different architectures (and different use-cases)

- 1 Encoder-Decoder (original article, T5, ...)
- 2 Encoder Only (BERT)
- 3 Decoder Only (GPT)



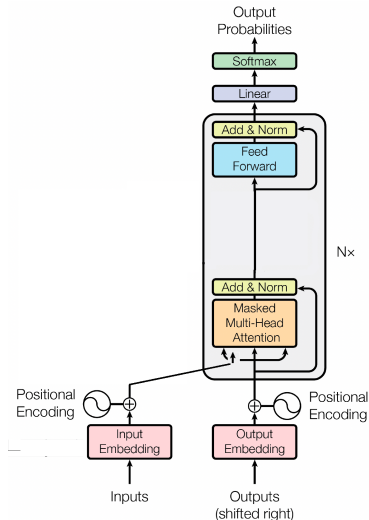
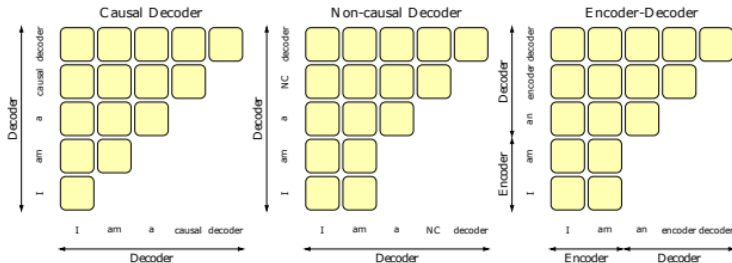
Different architectures (and different use-cases)

- 1 Encoder-Decoder (original article, T5, ...)
- 2 Encoder Only (BERT)
- 3 Decoder Only (GPT)



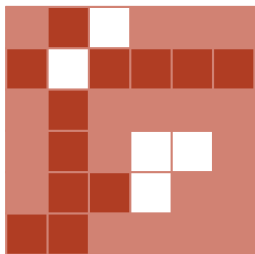
Different architectures (and different use-cases)

- 1 Encoder-Decoder (original article, T5, ...)
- 2 Encoder Only (BERT)
- 3 Decoder Only (GPT)



Masking attention computation

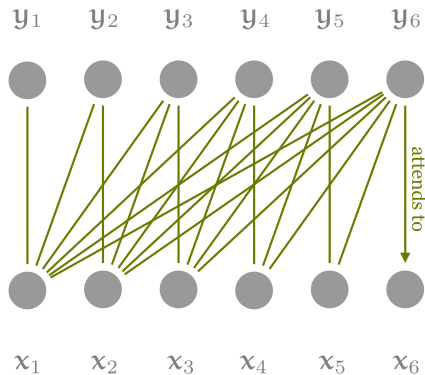
Constraint on the attention to improve prediction



raw attention weights



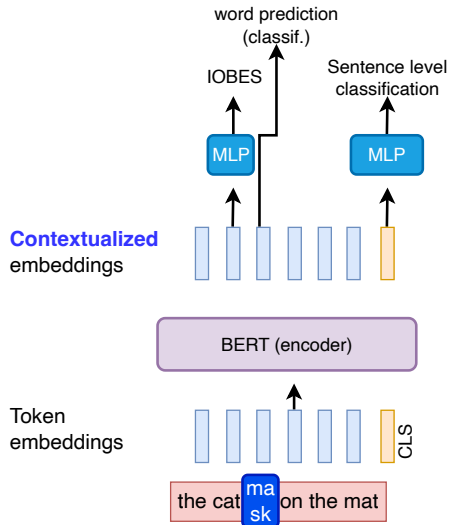
mask



PRE-TRAINING / TRAINING TASKS

From Word2Vec to Masked-LM (MLM)

- Masking token (+masking attn s.t. mask has no influence)
- Next token prediction (GPT)





Next sentence

- Learn to aggregate... cheap manner

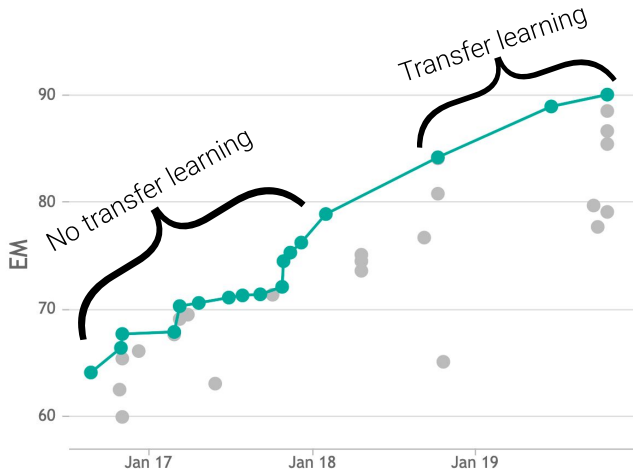
On chercher à savoir si une phrase en suit une autre

Input = [CLS] the man went to [MASK] store [SEP] he bought a gallon
[MASK] milk [SEP]
Label = IsNext

Input = [CLS] the man [MASK] to the store [SEP] penguin [MASK] are
flight ##less birds [SEP]
Label = NotNext

T5 architecture: generalizing gen. AI

- Refining a generative model to answer different tasks ... T5 :)
- <https://colinraffel.com/talks/mila2020transfer.pdf>

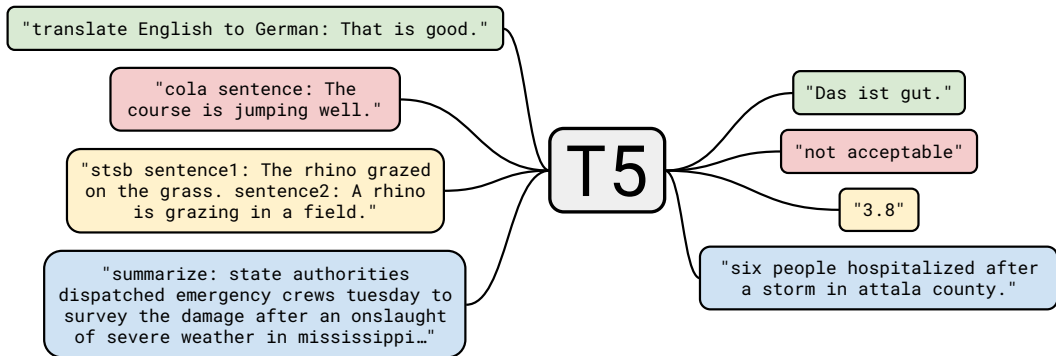


Source: <https://paperswithcode.com/sota/question-answering-on-squad11-dev>



T5 architecture: generalizing gen. AI

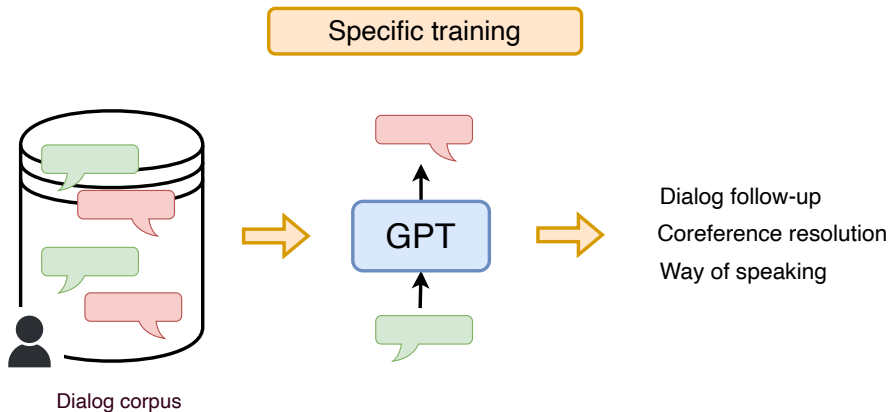
- Refining a generative model to answer different tasks ... T5 :)
- <https://colinraffel.com/talks/mila2020transfer.pdf>





The Ingredients of chatGPT

2. Dialogue Tracking

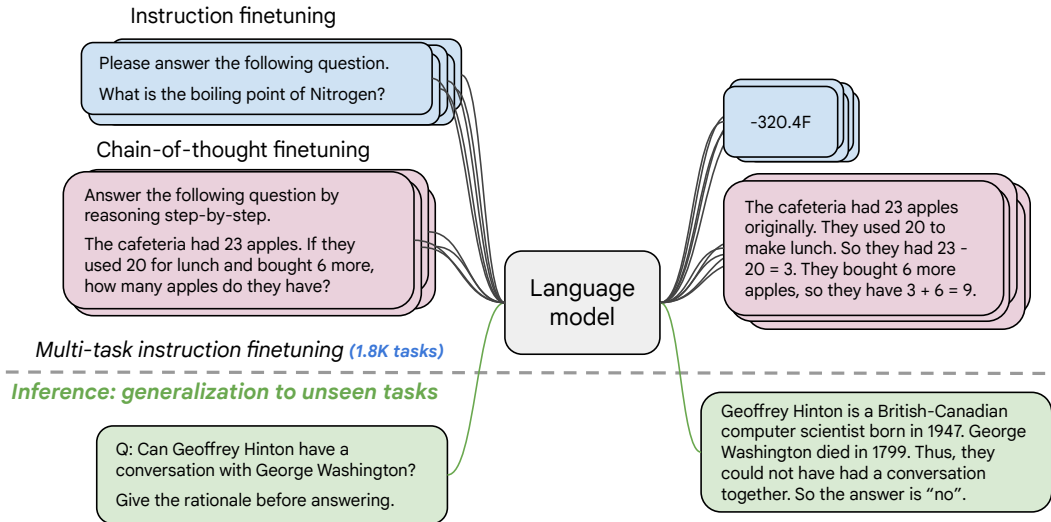


■ **Very clean** data

Data generated/validated/ranked by humans

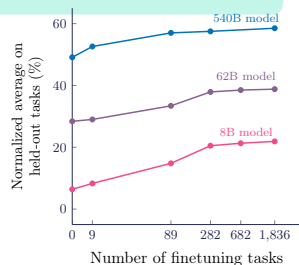
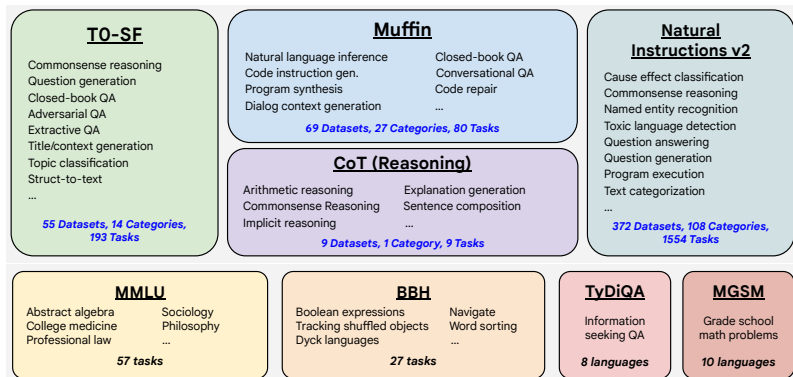
The Ingredients of chatGPT

3. Fine-tuning on different ($\pm$) complex reasoning tasks

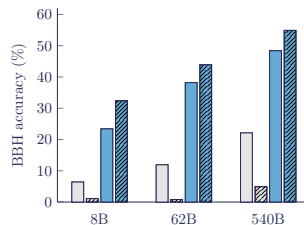


The Ingredients of chatGPT

3. Fine-tuning on different ($\pm$) complex reasoning tasks



- PaLM: Zero-shot
- ▨ PaLM: Zero-shot + CoT
- Flan-PaLM: Zero-shot
- ▩ Flan-PaLM: Zero-shot + CoT

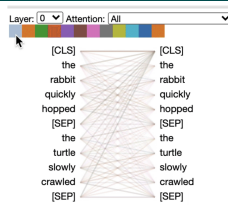


Scaling Instruction-Finetuned Language Models, Chung et al., JMLR 2024

EXPLOITATION

Layer visualization

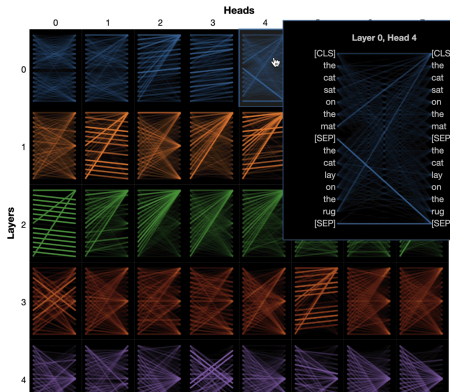
- BERT viz: <https://github.com/jessevig/bertviz>



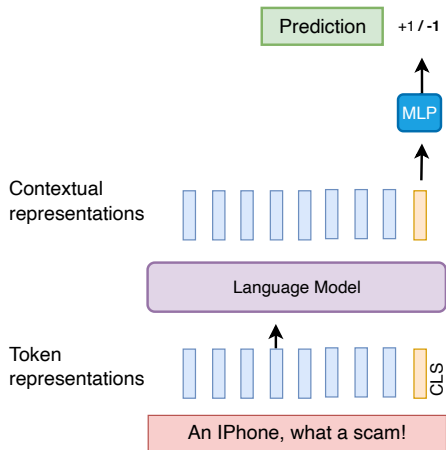
Model View

The *model view* shows a bird's-eye view of attention across all layers and heads.

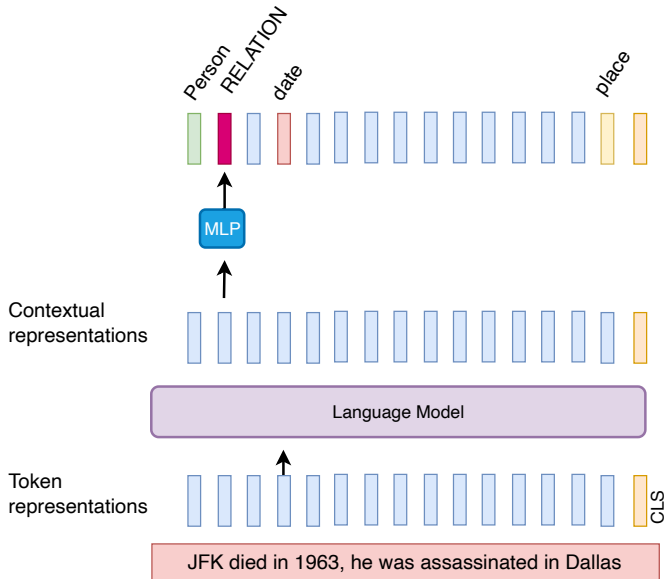
Try out the model view in the [Interactive Colab Tutorial](#) (all visualizations pre-loaded).



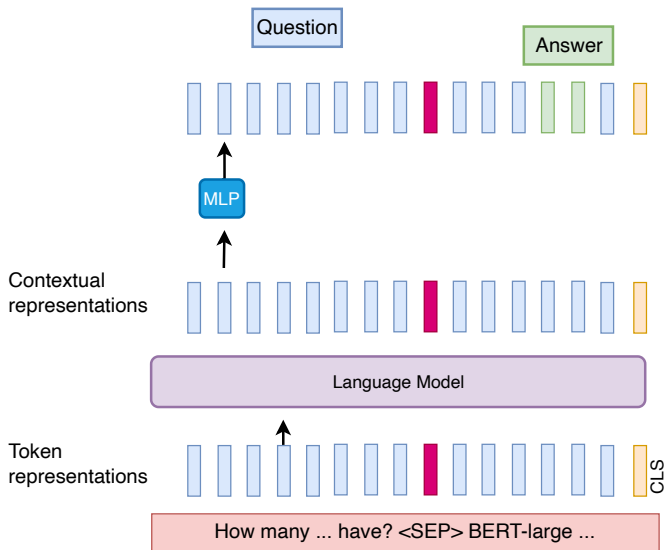
Example BERT



Example BERT

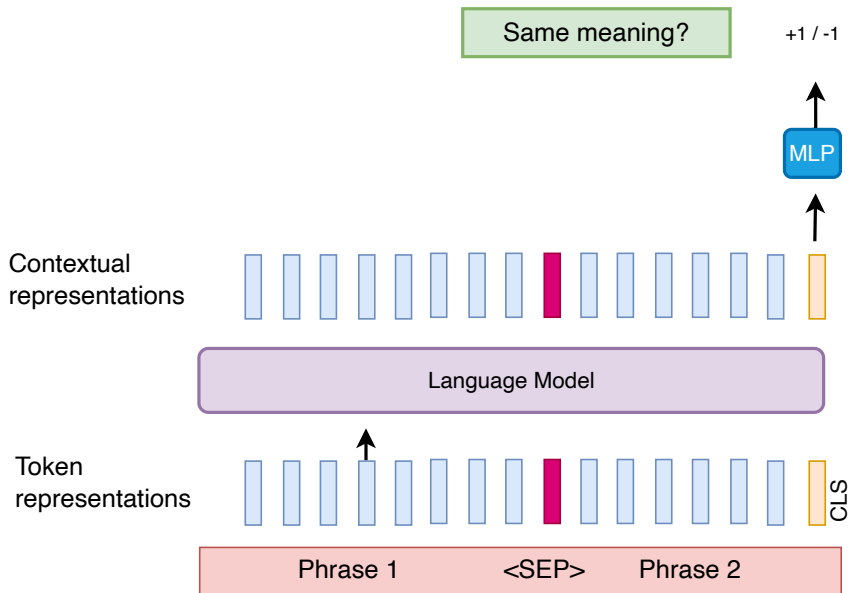


Example BERT



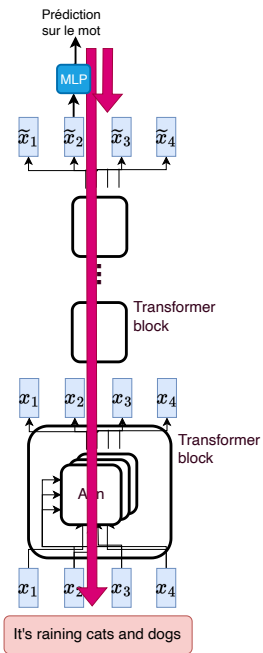


Example BERT

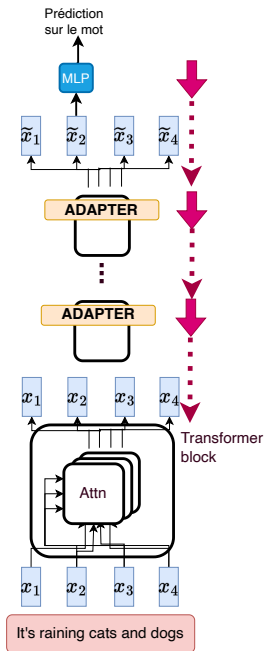




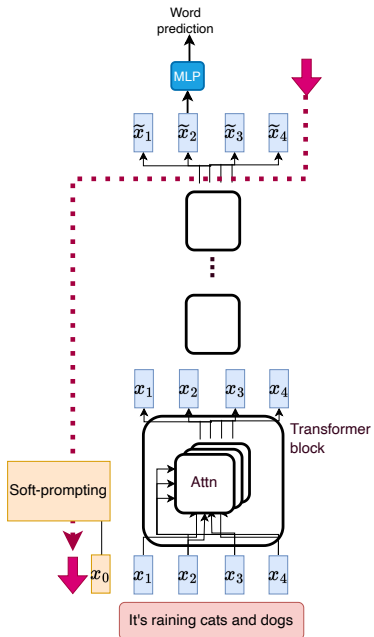
Different fine-tuning strategies



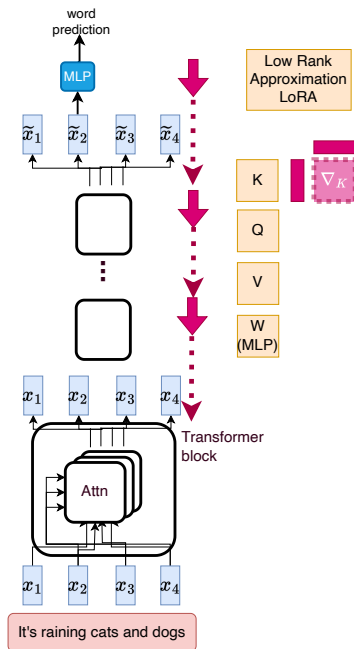
Different fine-tuning strategies



Different fine-tuning strategies



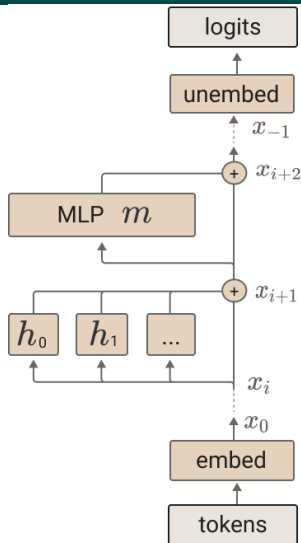
Different fine-tuning strategies





Care of novelty: improvements are not guaranteed

- New ways to see the transformer architecture (Anthropic)

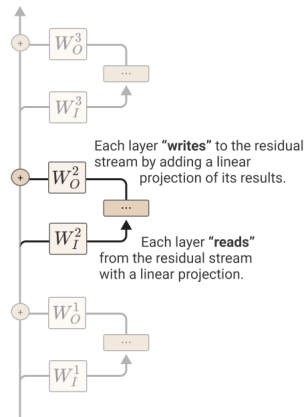


<https://transformer-circuits.pub/2021/framework/index.html>



Care of novelty: improvements are not guaranteed

- New ways to see the transformer architecture (Anthropic)



<https://transformer-circuits.pub/2021/framework/index.html>



Care of novelty: improvments are not guaranted

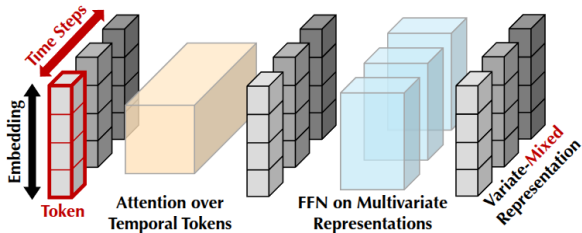
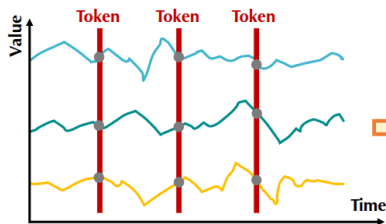
- New ways to see the transformer architecture (Anthropic)
- Time series

Are Transformers Effective for Time Series Forecasting? Zeng et al. 2022



Learning tokens... For every data modalities

■ Times series



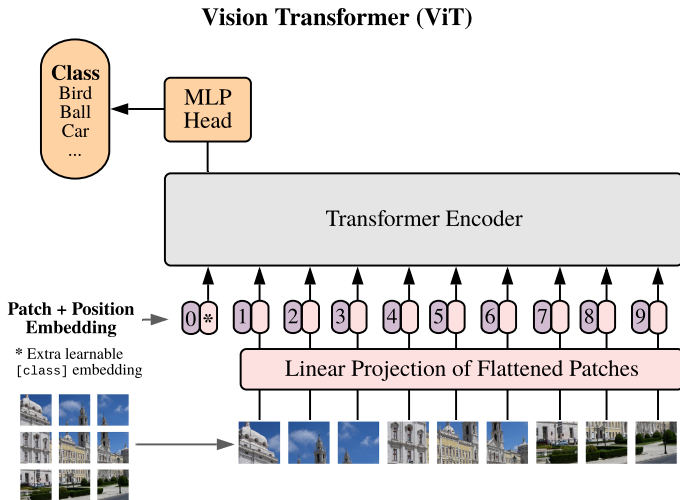
■ Images

■ ...

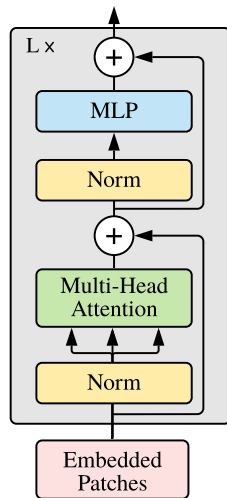
⇒ pre-training, masking, data-augmentation, ...

Learning tokens... For every data modalities

- Times series
- Images



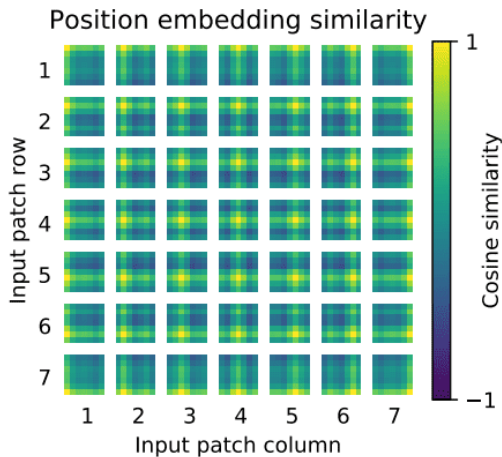
Transformer Encoder





Learning tokens... For every data modalities

- Times series
- Images



■ ...

⇒ pre-training, masking, data-augmentation, ...